Experimentation analysis of VQC and QSVM on sentence classification in quantum paradigm

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***Abstract*— The quantum computing made complex computations may be executed at light speed, superpositions allow for parallelism, and massive quantities of data can be efficiently processed. Furthermore, researchers and engineers are quickly becoming interested in natural language processing (NLP) as a topic for developing more comprehensive computational models of NLP. Therefore, there is hope for future progress in NLP efforts that use quantum technology, especially in emotion classification. In this study, we use quantum computing models to delve deeply into two sentiment categorization approaches. To be more specific, the effectiveness of the models is evaluated by looking at VQC and QSVM. The analyzed models are evaluated by simulations in the Qiskit simulator and the real-time quantum computer Qiskit_ibm_osaka. Results demonstrate the importance of the experiment and validate it using positive predicted value, sensitivity and F1-Score as performance metrices.**

Keywords*— *VQC, QSVM, sentence classification, Natural Language processing.

I. INTRODUCTION

Natural Language Processing (NLP) is a field of Artificial Intelligence that is constantly advancing and holds great significance in both theory and practice [6]. Some even argue that we are on the verge of reaching a point where machines can pass the Turing test. These models, typically relying on neural networks, operate by acquiring the ability to predict the likelihood of certain words based on their context. Textual data used for training is extracted from extensive text corpora [7]. The probability distribution that were indulged to learn extracts the patterns of word cooccurrence among the available content.

It is becoming more common to see natural language processing (NLP) technology integrated into several aspects of daily life, such as search engines, personal assistants etc., [8]. In spite of the fact that conventional natural language processing models have been successful in both language interpretation and language creation, deep neural networks are the only ones that can be found behind the hood of these models. Deep neural networks are notoriously criticized for being black boxes [9].

A small number of QNLP specialists, inspired by the striking resemblance of Quantum theory, assert that natural language processing issues are inherently "Quantum native" and look forward to a future when more powerful quantum hardware is accessible, promising a large-scale quantum computing advantage [10]. Since quantum and NLP both share common vector spaces - which are more efficiently represented by quantum hardware—QNLP models may outperform conventional direct computation approaches by a factor of four or five. Experimental results on existing quantum devices have shown a computational advantage over certain natural language processing (NLP) classical approaches, according to certain recent studies [11]. Our investigations build upon those first tests to demonstrate the future promise of QNLP approaches relative to some traditional ones, as measured by improved prediction accuracy values.

In this research work, sentence classification models are experimented using the quantum machine learning methods such as Quantum SVM and (QSVM) Variational Quantum Classifier (VQC). Preprocessing methods such as data encoding to be a quantum states are defined using DisCoCat and the models are evaluated under three different scenarios using different training and testing sets. The rest of this research paper is organized as follows: Section 2 comprises of the related works that are carried out on sentence classification using quantum machine learning models, section 3 comprises of the problem definition and description of data encoding. Section 4 deals with the experimentation results and analysis and section 5 concludes the research work and along with the description of future work.

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II. LITERATURE SURVEY

Presented in a more contemporary light, DisCoCat is language model which holds the interpretation of tensor network in the literature. A recent uptick in tensor network use in language modelling may be attributed to the need to create AI systems that can be understood and utilized by humans [16]. Graphs having tensors attached to their vertices are called tensor networks. The arity of a vertex is the total number of ends connected among, which stands for the tensor's indexes. Tensor contractions, represented by edges, are the result of summing across the range of the indices that are united by the edge (Eisert summation). Tensor networks are, intuitively, just multi-linear maps compressed into a single representation. In addition to their formal attractiveness as rigorous algebraic tools, it is used to represent probability distributions. There have been a lot of fascinating new developments in the last five years as a result of the obvious conceptual merging of QC with AI, particularly the branch of AI called machine learning (ML).

The study suggests a QNLP hybrid transfer learning model that may be used for binary classification. For the classification, it makes use of two fine-tuned pre-trained models: BERT and ELECTRA. In a recent study by Torlai and Melko [12], this method was put to the test on NISQ machines, which are noisy intermediate-scale quantum computers. It uses parametrized variational circuits [13] for classification and quantum encodings [14] to transform classical input into a quantum state, while also taking use of conventional pre-trained language characteristics for their resilience and advantages. Binary classification on acceptability judgments is the goal of the VQC algorithms here. This is an important problem in computational linguistics and theoretical linguistics, as well as in computation, which has recently become very popular in natural language processing [15].

III. PROBLEM DEFINITION, QSVM AND VQC ARCHITECTURES

In this section the problem statement is defined along with the description of QSVM and VQC models.

3.1 Problem Definition

We will now describe the text categorization issue that this research aims to solve: Here you may find a compilation of SMS texts. There is a character limit of 175 for each message. It includes both standard and unique characters. We need to construct a system that can automatically determine whether any of the 5572 messages in our dataset are spam. So, this is a binary classification scenario, while the same theory and a very similar network design may be used to expand it to multi-class instances. In order to make classical-quantum transfer learning via dressed quantum circuits viable, the BERT model is fine-tuned for text categorization. In Figure 1 we can see the big picture of our problem's network architecture.

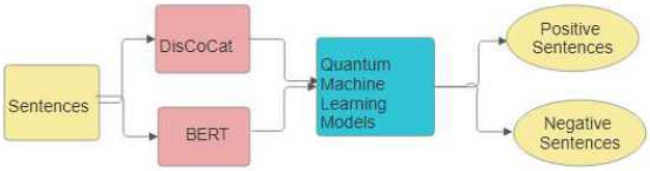


Figure 1: The Experimentation flow

3.2 Quantum Support Vector Machine

Using a least squares variant of the traditional support vector machine (SVM) [2], the QSVM was shown in [1]. The aim of SVMs is to classify vectors into one of two classes, where Y_K which could be either +1 or -1, liable on the class (X_k), given m training data points of the type (X_K,Y_K) where K ranges from 1 to m. A linear equation expressing LS-SVM is possible:

$$A\begin{pmatrix} b \\ a \end{pmatrix} = \begin{pmatrix} 0 & \mathrm{I}_T \\ 1 & K + y^{-1} \end{pmatrix} \times \begin{pmatrix} b \\ a \end{pmatrix} = \begin{pmatrix} 0 \\ Y \end{pmatrix} \quad (1)$$

where A refers the input matrix such that A= $\begin{pmatrix} 0 & \mathrm{I}_T \\ 1 & K + y^{-1} \end{pmatrix}$.

In QSVM and its subsequent iterations, it is anticipated that the input matrix A will exhibit sparsity and a well-organized structure. From ML point of view, it is not anticipated that the input matrices would be sparse or well-structured, hence limiting the potential uses of the HHL method. Singular value decomposition is a crucial technique for solving linear equations. The AE-QSVM [3] is trained using the quantum singular value decomposition (QSVD) in this study. In other words, Equation (1) will be solved using the QSVD method.

The singular value of square matrix A is equivalent to its eigenvalue. The decomposition of matrix A into its singular values using the eigenvalues and the associated eigenvectors. The solution A may be obtained for the Moore-Penrose pseudoinverse if the matrix A is unique, meaning that the nonzero eigenvalues are merely inverted.

The quantum linear systems A|b,a> =|0,Y> are solved using QSVD [3]. Initially, the QRAM [4] is used to prepare |0,Y>. Furthermore, a quantum state is established for both the labeled dataset and the unlabeled dataset in order to categorize the new sample.

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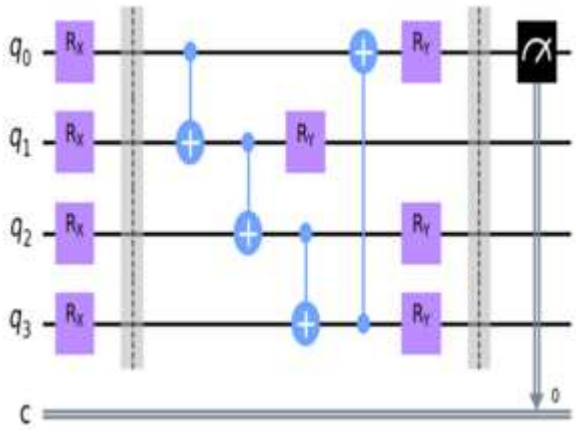


Figure 2: Generic QSVM circuit

// Dataset Preprocessing
Step 1: Load the sentiment dataset
Step 2: Tokenize and encode the text data into numerical format
Step 3: Split the data into training and testing sets

// Mapping the feature sets
Step 4**: Map classical data to quantum states using ZZfeaturemap**

// QSVM model
Step 5: Initialize the QSVM with the defined feature map
Step 6: Set the parameters for the SVM

// QSVM model Training
Step 7: Input the training data to the QSVM model
Step 8: Train the model to find the optimal hyperplane for classification

// QSVM model Testing
Step 9: Input the testing data to the trained QSVM model
Step 10: Evaluate the performance of the model

Algorithm 1: Pseudocode for QSVM on sentiment Classification

3.3 Variational Quantum Classifier

The simplest quantum gates on the market allow us to do computational source. For this reason, our VQC includes a variety of one-qubit gates, such as the Hadamard gate H and rotating gates, as well as entangling gates, such as the controlled-NOT and controlled-Z. The Appendix provides an introduction to each of them and includes their explicit form in addition to the names used to refer to them.

We use quantum systems that are initiated in the $|0>^{\otimes} n$ state for our circuits. The rationale for this is that states with a range of $|00 \ldots 0>$ may be easily constructed. According to the Solovay-Kitaev theorem [5], almost all unitary gates that operate on n qubits may be reduced to a collection of these fundamental gates, with a certain degree of accuracy denoted as ε.

The accuracy of the categorization is determined by the angles θ_1 to θ_l, where l is the total number of rotations. We use a conventional approach to complete the second portion of the VQC. Using the final state of the quantum circuit, which is a function of θ_i, we first calculate a function. Next, using a strategy similar to that of the ANN, the algorithm endeavors to reduce this function.

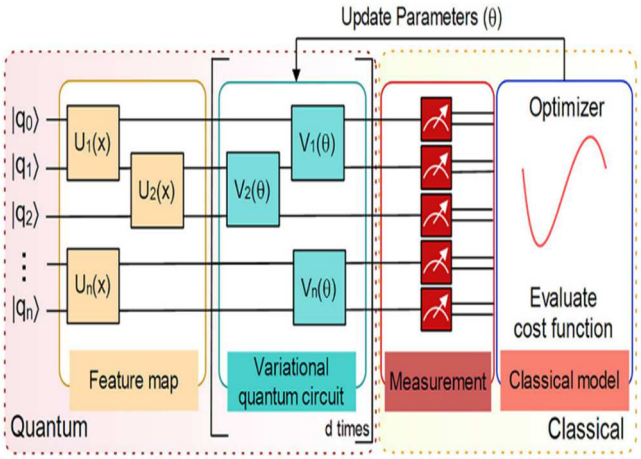


Figure 3: VQC

IV. EXPERIMENTAL ANALYSIS

The lambeq toolkit is used to perform QNLP on our sentiment analysis dataset. A virtual machine of type n1-highmem-8, fifty-two gigabytes of RAM, eight virtual CPUs, and Linux make up the experimental configuration. There has been no change to the hardware since the exploratory work in [18]. The experiment's dataset consisted of a collection of simple subjective terms pertaining to the restaurant industry. These sentences, which include both positive and negative assertions, are constructed using a total of 29 vocabulary. They are four to five words long. The order of the sets are stated as (Training%, Testing %) and the sets are defined as: Set -1 (80, 20), Set -2 (70, 30), and Set -3 (60, 40) are the three divisions that pertain to the training and testing splits. Furthermore, the information was split into ten distinct situations, each with its own unique combination of positive and negative remarks. The considered performance metrices are positive predicted value (PPV), sensitivity and F1-Score.

Scenario	VQC			QSVM		
	PPV	Sensitivity	F1-Score	PPV	Sensitivity	F1-Score
1	100 %	99.56 %	100%	98.61%	97.61 %	98.11 %
2	99.64%	98.26 %	98.94 %	97.11%	95.94 %	96.52 %
3	98.32%	97.80 %	98.06 %	97.90%	95.21 %	96.54 %
4	98.56%	97.20 %	97.88 %	97.05%	94.27 %	95.64 %
5	100 %	100%	100%	100 %	100%	100%
6	100 %	98.76 %	99.37 %	98.40%	95.73 %	97.05 %
7	100 %	100%	100%	98.29%	94.75 %	96.49 %
8	97.16%	96.10 %	96.63 %	96.42%	94.12 %	95.26 %
9	100 %	100%	100%	97.92%	96.32 %	97.11 %

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10	100 %	100%	100%	98.83%	96.68 %	97.74 %

Table 1 tabulated the results of VQC and QSVM under ten scenarios on split set 1. On interpreting the results of two quantum machine learning models, on average positive predicted value VQC dominate QSVM in 1.3% of efficiency. On interpreting the results of sensitivity VQC dominate QSVM in 2.8% of efficiency. And on interpreting the results of F-1 Score VQC dominate QSVM in 2.1% of efficiency. Figure 4 gives the graphical interpretation of results from Table 1.

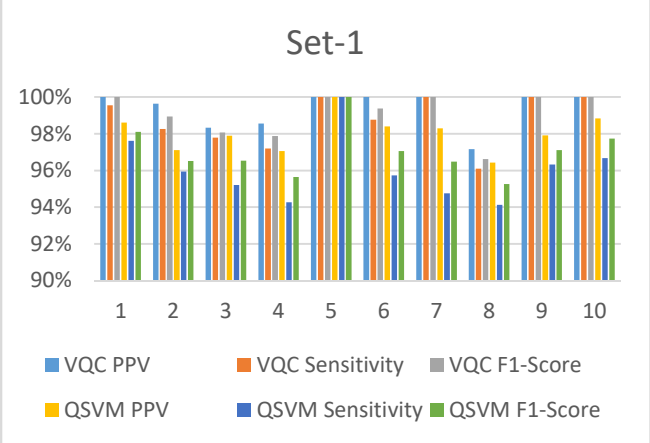


Figure 4: Results of VQC and QSVM on set 1

Table 2 tabulated the results of VQC and QSVM under ten scenarios on split set 2. On interpreting the results of two quantum machine learning models, on average positive predicted value VQC dominate QSVM in 1.2% of efficiency. On interpreting the results of sensitivity VQC dominate QSVM in 3.3% of efficiency. And on interpreting the results of F-1 Score VQC dominate QSVM in 3.2% of efficiency. Figure 5 gives the graphical interpretation of results from Table 2.

Scenario	**VQC**			**QSVM**		
	PPV	**Sensitivity**	**F1-Score**	**PPV**	**Sensitivity**	**F1-Score**
1	99.86%	99.88 %	99.87 %	99.50%	95.99 %	97.71 %
2	98.76%	98.68 %	98.72 %	96.53%	95.77 %	96.15 %
3	98.23%	97.87 %	98.05 %	96.49%	96.10 %	96.30 %
4	97.64%	98.30 %	97.97 %	96.55%	94.46 %	95.49 %
5	100 %	100%	100%	99.54%	97.06 %	98.29 %
6	100 %	100%	100%	98.92%	97.10 %	98.00 %
7	100 %	100%	100%	98.63%	96.24 %	97.42 %
8	96.87%	97.04 %	96.96 %	97.03%	93.78 %	95.37 %
9	100 %	100%	100%	98.76%	96.97 %	97.85 %
10	100 %	100%	100%	97.63%	96.35 %	96.99 %

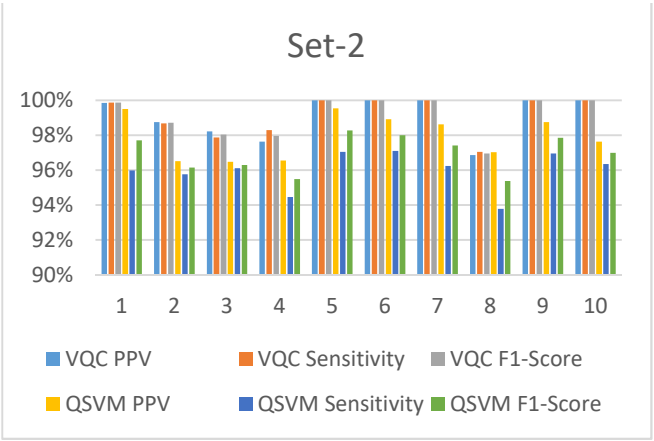


Figure 5: Results of VQC and QSVM on set 2

Table 3: Results of QSVM and QLSTM for different scenarios on Set - 3

Scenario	**VQC**			**QSVM**		
	PPV	**Sensitivity**	**F1-Score**	**PPV**	**Sensitivity**	**F1-Score**
1	100 %	100%	100%	98.04%	96.84 %	97.44 %
2	100 %	100%	100%	98.72%	95.16 %	96.91 %
3	99.85%	99.90 %	99.88 %	96.57%	95.99 %	96.28 %
4	100 %	100%	100%	95.98%	95.59 %	95.79 %
5	99.11%	99.12 %	99.11 %	98.95%	98.15 %	98.55 %
6	100 %	100%	100%	98.94%	98.59 %	98.76 %
7	99.55%	99.47 %	99.51 %	97.76%	96.38 %	97.06 %
8	98.56%	99.16 %	98.86 %	94.82%	94.92 %	94.87 %
9	99.21%	99.59 %	99.40 %	97.91%	96.19 %	97.05 %
10	100 %	100%	100%	97.67%	98.16 %	97.91 %

Table 3 tabulated the results of VQC and QSVM under ten scenarios on split set 3. On interpreting the results of two quantum machine learning models, on average positive predicted value VQC dominates QSVM in 2.1% of efficiency. On interpreting the results of sensitivity VQC dominate QSVM in 3.2% of efficiency. And on interpreting the results of F-1 Score VQC dominate QSVM in 2.7% of efficiency. Figure 6 gives the graphical interpretation of results from Table 3.

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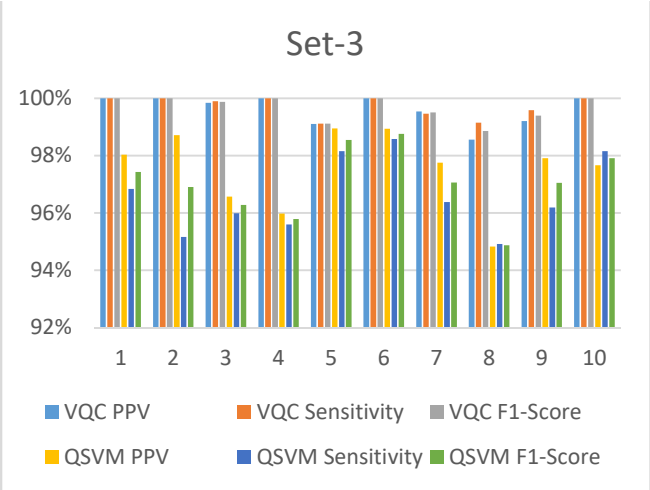


Figure 6: Results of VQC and QSVM on set 3

V. CONCLUSION

In this study, we examine the experimental results of using VQC and QSVM for sentiment categorization. The dataset included in this study comes from a restaurant domain, and its tweets have been pre-processed using several text mining techniques, including stop word removal. There are ten distinct situations with varying phraseologies, and the datasets are divided into three groups according to the quantity of samples used for training and testing. The VQC performs better than the QSVM in several respects, including positive predicted value, sensitivity and F1-Score. Adding other quantum models for sentiment categorization may expand this study in the future.

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